

Properties of equality and congruence



- 1 a $b + c$ b bc c AB
- 2 The multiplication property of equality.
- 3 a The symmetric property of equality. b The addition property of equality.
c The transitive property of congruence.
- 4 a No b No c Yes d No
- 5 a Yes b Yes c No d Yes
- 6 a No b No c No d Yes
- 7 a The transitive property of equality. b The multiplication property of equality.
c The addition property of equality.
- 8 a No b Yes c No d No
e Yes f No



Additional questions

9

Property of Equality	Example
Reflexive property of equality	$a = a$
Symmetric property of equality	If $a = b$, then $b = a$.
Transitive property of equality	If $a = b$ and $b = c$, then $a = c$.
Addition property of equality	If $a = b$, then $a + c = b + c$.
Subtraction property of equality	If $a = b$, then $a - c = b - c$.
Multiplication property of equality	If $a = b$, then $a \times c = b \times c$.
Division property of equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.
Substitution property of equality	If $a = b$, then b may be substituted for a in any expression containing a .

10

- a Addition property of equality
- b Converse of multiplication property
- c Converse of addition property of equality
- d Converse of subtraction property of equality

11

- a Converse of transitive property of equality
- b Substitution property of equality